ESTIMATION OF ROOFTOP SOLAR PV SYSTEM (1 - 20 kW)

Date: 22-Sep-2026

Site: Case B - Statutory Capped Customer | Location: Mysuru, Karnataka | Grid-Tied Solar Engineering Dossier

PLANT CAPACITY	ANNUAL YIELD	CAPITAL COST	PAYBACK PERIOD
550W Mono	Specific Yield	12-Item BOM	25Y Lifecycle
20.00 kW	**32,266 kWh**	**Rs. 887,800**	**4Y 5M**
20.35 kWp Installed	CUF 18.1% | PR 75%	Rs. 8.88 Lakhs	Profit: Rs. 95.50L

[SITE] Customer & Site Engineering Parameters

Verified project configuration, contact details, and sizing pathway

Company / Customer Name	Case B - Statutory Capped Customer
Contact Mobile / Phone	+91 97777 22222
Location / Address	Mysuru, Karnataka
Contact Email Address	case_b@voltroof.com
Design Path Selected	Rooftop Area Analysis
Approved Sanctioned Load (CD)	25 kW (KERC / ESCOM Tariff Order)
Rooftop Area Verification	Sufficient Rooftop Area (+1400 sq.ft surplus)
System Target Capacity	20.00 kW (Single LT Installation)
Total Installed Array Rating	20.35 kWp
Solar PV Module Rating	550W Mono PERC / TOPCon Half-Cut
Total Module Count	37 Nos.

[METRICS] System Performance & Economic Evaluation

25-Year performance metrics and grid benchmark comparisons

Total Rooftop Area	3000.00 Sq.Feet (278.71 sq.m)
Utilized Usable Rooftop Area	3000 Sq.Feet (278.71 sq.m)
First Year Annual Generation	32,265.73 kWh
25-Year Cumulative Generation	716,875.60 kWh
Estimated Payback Period	4 Years & 5 Months
Post-Payback Profitability (25 Yrs)	Rs. 9,549,571.78 (INR 95.50 Lakhs)
Alternative FD 6.5% Benchmark (25 Yrs)	Rs. 4,286,031.27 (INR 42.86 Lakhs)

ENVIRONMENTAL & SUSTAINABILITY IMPACT (25-YEAR LIFECYCLE)

Carbon Offset: 587 Tonnes CO2 | Tree Equivalence: 939 Mature Trees | Green Energy: 100% Clean LT Power

Based on standard Central Electricity Authority (CEA) grid emission and forestry carbon sequestration benchmarks.

20 kW Reached For LT Limited 20kW as per KERC/ESCOM Regulations

Statutory ceiling enforced under KERC SRTPV Regulations. System automatically sized to maximum allowable 20.0 kW.

[ARRAY] Building 1 Layout & Rooftop Array Sizing

Building structural envelope, array capacity, and module layout

Building / Array Identification	Building 1 (Rooftop Array)
Gross Rooftop Area	3000.00 Sq.Feet (278.71 sq.m)
Utilized Effective Rooftop Area	3000 Sq.Feet (278.71 sq.m)
Building Plant Target Capacity	20.00 kW
Solar PV Module Rating	550 W Mono-PERC (Mono-facial)
Number of Solar Modules	37 Nos.
String Sizing Architecture	18 - 20 Panels per MPPT String (Optimal Voltage Band)
Selected Site Inverter Rating	15 kW, 415V 3-Ph MPPT

[INVERTER] Selected Grid-Tied Inverter Specifications (15 kW)

Core inverter rating (70% derated selection), MPPT topology & models

Selected Inverter Rating	15 kW Grid-Tied Inverter (Three-Phase)
Certified Top-Tier OEM Models	Sungrow SG15RT, UTL On-Grid
Derated AC Design Power (70% Factor)	14.24 kW AC (from 20.00 kWp DC)
AC Output Voltage & Grid Phase	415V, 3-Phase, 50 Hz AC
MPPT Operating Topology	Dual MPPT Architecture (>99.5% MPPT Tracking Efficiency)
Enclosure & Grid Protection	IP65 Weatherproof Enclosure with Anti-Islanding (IEC 62116)
Inverter to ACDB Cable	3.5Cx25 Sq.mm
DC Array Isolator Switch	63A, MCB Type, 4P 415V

[MCB] Selected MCB & Switchgear Protection Specifications

Breaker sizing, trip characteristics, and surge protection devices

Incomer Overcurrent Protection	Thermal-Magnetic MCB / MCCB with Manual Isolation
Breaker Current Rating	25 Amps
Rated Inverter Operating Current	19.81 Amps
Trip Curve & Breaking Capacity	Type C Curve / 10 kA Breaking Capacity (IS/IEC 60898-1)
Surge Protection Device (SPD)	Type II AC SPD (40 kA, 415V / 230V) with Remote Signalling

[CABLES] Electrical Cable Sizing & Interconnection Schedule

Cable conductor specifications, insulation, and voltage drop compliance

Solar PV DC Array Cable	4 / 6 sq.mm UV-Resistant Solar DC Cable (1.5 kV, EN 50618)
Inverter AC Output Cable	3.5Cx25 Sq.mm (Cu / XLPE / Armoured)
ACDB to Utility Grid Cable	3.5Cx6 Sq.mm (Al / XLPE / Armoured)
Earth Grounding Conductor	16 sq.mm Cu / GI Earth Conductor connected to Dual Chemical Pits
Design Standard & Voltage Drop	Voltage Drop < 2% in accordance with IS 694 / IEC 60364-5-52

[ACDB] AC Distribution Board (ACDB 1 IN / 1 OUT) Specifications

Switchgear enclosure, busbar rating, and metering point

ACDB Enclosure Specification	IP65 Weatherproof Polycarbonate / Sheet Steel
Internal Busbar Rating	50A, 415V, 3PN Aluminium
Incomer & Outgoing Terminations	Cable Glands with Compression Lugs & Dual Earthing
Interconnection Distance	20.0 Meters (ACDB to Utility Net-Meter)

[BOM] Turnkey Plant Bill of Materials (BOM)

Complete 12-item engineering specification and cost breakdown

20 kW Reached For LT Limited 20kW as per KERC/ESCOM Regulations

Sl.	Component / Material Description	Engineering Specification	Qty	Amount (INR)
01	Solar PV modules	550W Mono-PERC (Mono-facial)	37 Nos.	Rs. 370,000
02	On-grid inverter	15 kW, three-phase (Sungrow SG15RT or equiv.)	1 No.	Rs. 157,500
03	Rooftop mounting structure	HDGI, module-compatible	1 Set	Rs. 95,000
04	DC solar cable	6 sq.mm copper, PV-rated	220 m	Rs. 18,700
05	AC cable	35 sq.mm copper	110 m	Rs. 49,500
06	MC4 connectors	PV-rated connector pairs	37 pairs	Rs. 11,100
07	DCDB	Isolator + fuse + Type-II DC SPD	1 No.	Rs. 18,000
08	ACDB	25A MCB + 40A RCCB + Type-II AC SPD	1 No.	Rs. 18,000
09	Earthing kit	Earth electrodes & accessories (IS 3043 - 3 Pits)	3 Nos.	Rs. 9,000
10	Lightning arrester	Standard rooftop PV type	1 No.	Rs. 8,000
11	Installation accessories	Lugs, ferrules, clamps, conduit, glands, etc.	1 Lot	Rs. 38,000
12	Installation & commissioning	Labour and commissioning	1 Job	Rs. 95,000
		TOTAL PLANT-SIDE CAPITAL COST (12-ITEM BOM)		**Rs. 887,800.00**

[UTILITY] ESCOM / Utility Statutory Approvals & Grid Budget

Official DISCOM statutory filing fees and net-metering charges

Application & Registration Fee	Rs. 1,000 + GST (DISCOM portal filing)
Facilitation & Technical Processing Fee	Rs. 2,000 + GST (Feasibility clearance)
PPA e-Sign Stamp Duty Charge	Rs. 11.8 (Electronic stamping)
Bidirectional Utility Net-Meter	Rs. 12,000 (DISCOM meter testing/procurement)
Refundable Security Deposit	Rs. 8,000 (Refundable utility escrow)

TOTAL CAPITAL INVESTMENT REQUIRED

Turnkey Plant BOM Cost: Rs. 887,800 | ESCOM Utility Budget: Rs. 23,012 | Grand Total Outlay: Rs. 910,812 (Rs. 9.11 Lakhs)

[FINANCE] 25-Year Financial Amortization Schedule

Annual cash flow model incorporating 1%/year degradation & tariff escalation

CAPITAL INVESTMENT (CAPEX)	PAYBACK PERIOD (RETURN)	25Y NET SAVINGS (PROFIT)	FD 6.5% BENCHMARK (BENCHMARK)
Rs. 887,800	**4Y 5M**	**Rs. 95.50L**	**Rs. 42.86L**
12-Item BOM Basis	Break-Even Year	Rs. 9,549,572	Alternative Asset

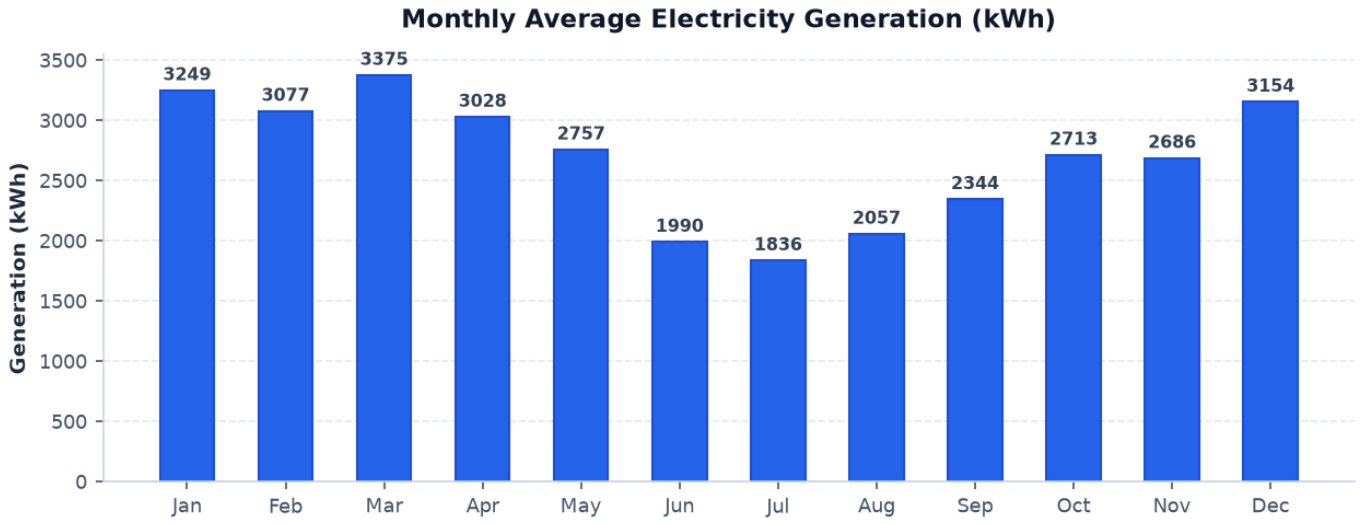
Yr	Gen (kWh)	Tariff (Rs/u)	Gross Rev (Rs)	AMC (Rs)	Net Annual (Rs)	Cumulative Net (Rs)
01	32,266	Rs. 8.05	259,739	12,000	247,739	247,739
02	31,943	Rs. 8.45	269,919	12,000	257,919	505,658
03	31,624	Rs. 8.87	280,502	12,000	268,502	774,160
04 *	**31,307**	**Rs. 9.32**	**291,785**	**12,000**	**279,785**	**1,053,945**
05	30,994	Rs. 9.78	303,125	12,000	291,125	1,345,069
06	30,684	Rs. 10.27	315,129	12,000	303,129	1,648,198
07	30,378	Rs. 10.78	327,470	12,000	315,470	1,963,668
08	30,074	Rs. 11.32	340,435	12,000	328,435	2,292,103
09	29,773	Rs. 11.89	354,001	12,000	342,001	2,634,104
10	29,475	Rs. 12.48	367,852	12,000	355,852	2,989,956
11	29,181	Rs. 13.11	382,557	12,000	370,557	3,360,513
12	28,889	Rs. 13.76	397,509	12,000	385,509	3,746,022
13	28,600	Rs. 14.45	413,268	12,000	401,268	4,147,290
14	28,314	Rs. 15.18	429,804	12,000	417,804	4,565,094
15	28,031	Rs. 15.93	446,529	12,000	434,529	4,999,624
16	27,750	Rs. 16.74	464,542	12,000	452,542	5,452,166
17	27,473	Rs. 17.57	482,699	12,000	470,699	5,922,865
18	27,198	Rs. 18.45	501,806	12,000	489,806	6,412,671
19	26,926	Rs. 19.37	521,560	12,000	509,560	6,922,231
20	26,657	Rs. 20.34	542,202	12,000	530,202	7,452,433
21	26,390	Rs. 21.36	563,698	12,000	551,698	8,004,132
22	26,126	Rs. 22.43	586,017	12,000	574,017	8,578,148
23	25,865	Rs. 23.55	609,125	12,000	597,125	9,175,274
24	25,607	Rs. 24.73	633,250	12,000	621,250	9,796,523
25	25,350	Rs. 25.96	658,098	12,000	646,098	10,442,622

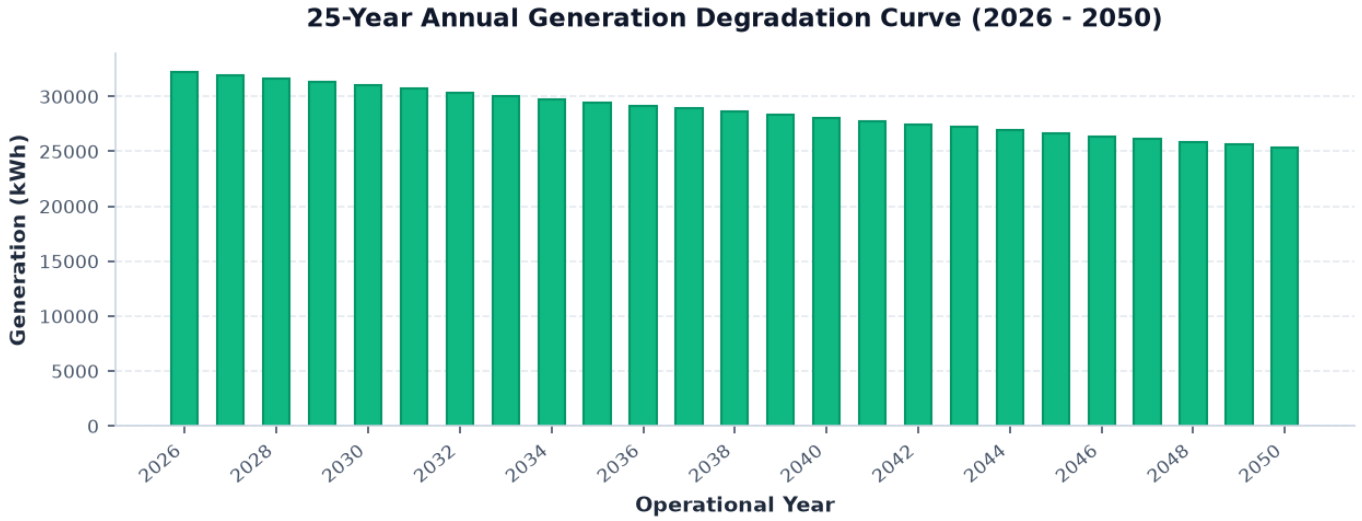
** Highlighted row indicates capital payback year. Annual maintenance contract (AMC) budgeted at Rs. 600/kW/year with 1% annual PV degradation.*

[ANALYTICS] Generation Analytics & Performance Curves

Monthly solar insolation yield and 25-year degradation profile

SPECIFIC YIELD (YIELD)	PERFORMANCE RATIO (PR)	CAPACITY UTIL. (CUF)	DEGRADATION RATE (DEGRADATION)
1613 kWh/kWp	**75.0%**	**18.1%**	**1.0% / year**
kWh/kWp/Year	IEC 61724 Standard	Annual Plant Factor	Tier 1 Linear Warranty





Simulated with regional GHI solar insolation database (3.26 to 5.40 kWh/m2/day). System performance calculated using PVsyst/MNRE benchmarks.

VoltRoof Designs
CLEAN ENERGY. BRIGHTER TOMORROW

ON-GRID SOLAR PV SYSTEM

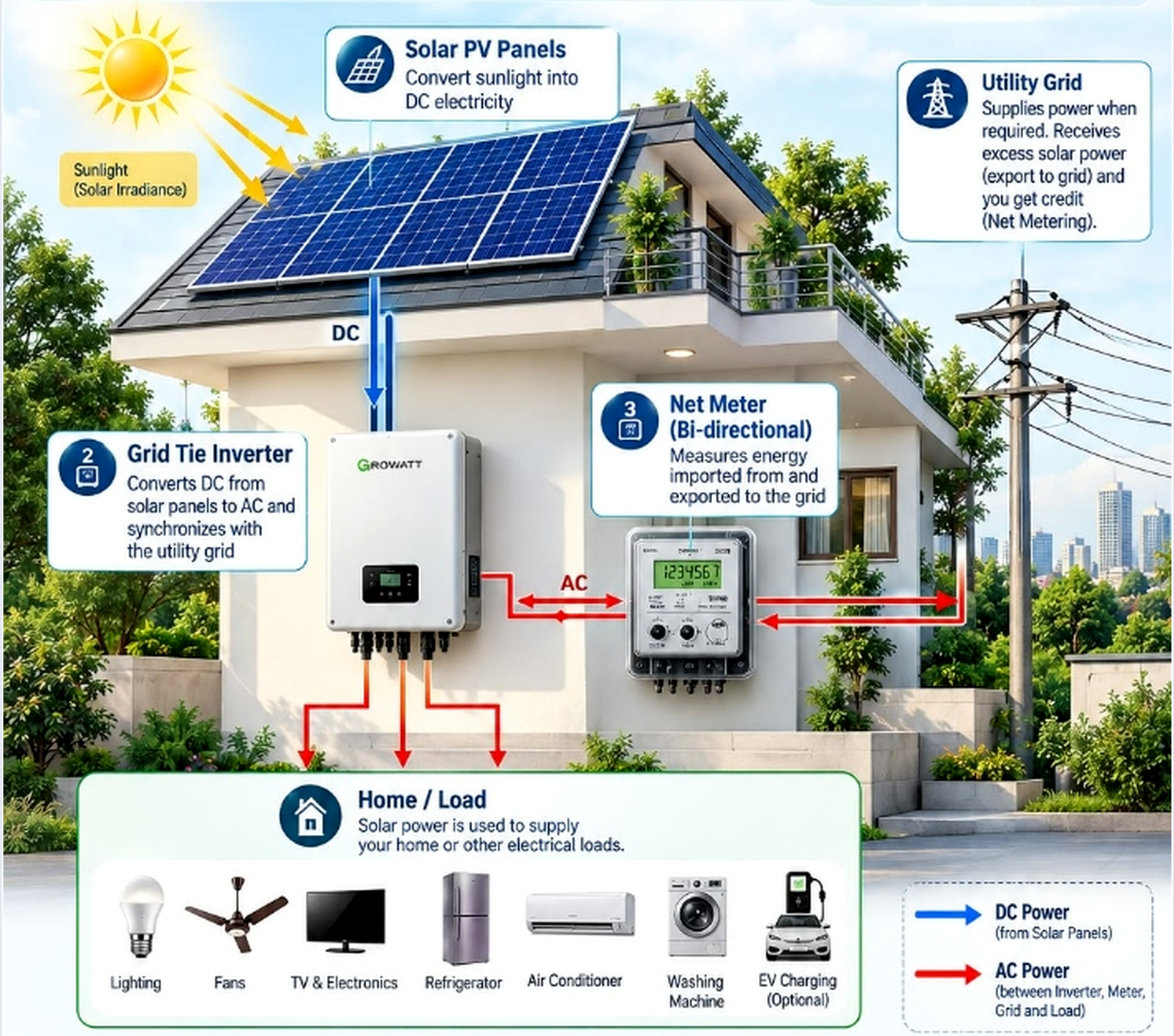
(Grid Connected Solar System)

GENERATE • CONSUME • EXPORT • SAVE

Harness The Sun For A Better Tomorrow

An on-grid solar PV system is connected to the utility grid. The solar power generated is used to meet your load demand. Any excess power is exported to the grid and you get credit through net metering. No battery is used in this system.

- Lower Electricity Bills
- Clean Energy
- Sustainable Future



Home / Load

Solar power is used to supply your home or other electrical loads.

Lighting | Fans | TV & Electronics | Refrigerator | Air Conditioner | Washing Machine | EV Charging (Optional)

DC Power (from Solar Panels)

AC Power (between Inverter, Meter, Grid and Load)

KEY ADVANTAGES

- Reduced Electricity Bills
- Lower Carbon Emissions
- No Battery Required
- Net Metering Benefits
- Low Maintenance
- Environment Friendly

VoltRoof Designs

Clean Energy Better Tomorrow